Emil Kasper

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Profile

Computer Science student at ETH Zurich, with a focus on reinforcement learning, robot learning, and human-AI interaction. Experience spanning machine-learning research, software engineering, and robotics.

Experience

IVIA Lab, ETH Zurich, Research Fellow (ETH SSRF) — Zurich, Switzerland
Supervised by Prof. Menna El-Assady. — July 2026 – Aug 2026 — 2 months
- Selected as one of 20 students from over 4,000 applicants for the ETH Student Summer Research Fellowship.
- Working on explainable AI and human-AI interaction.

Siemens, Software Engineer — Vienna, Austria
Aug 2025 – June 2026 — 11 months
- Contributing to a test simulation framework for the European Train Control System (ETCS).
- Implementing the autonomous train operation (ATO) mode in C# for test simulations.
- Presenting and pitching to stakeholders from Norway, Germany, Belgium, and Austria.

Institute for Visual Computing and Human-Centered Technology, TU Wien, Teaching Assistant — Vienna, Austria
Mar 2025 – June 2026 — 1 year 4 months
Course: Interface and Interaction Design.
- Guiding students through the lifecycle of user-centered product development.
- Providing feedback and grading for 45 students.

Machine Learning Research Unit, TU Wien, Undergraduate Researcher — Vienna, Austria
Supervised by Prof. Clemens Heitzinger. — Aug 2025 – Apr 2026 — 9 months
- Investigating the convergence and stability of Q-Learning and Speedy Q-Learning.
- Experimenting with PPO for robotic hand manipulation.

Education

MSc — **ETH Zurich**, Computer Science — Zurich, Switzerland
- Major in Machine Intelligence. — Sept 2026 – May 2028

BSc (Honors) — **TU Wien (Vienna University of Technology)**, Computer Science — Vienna, Austria
- Four-year honors degree with a minor in machine learning. — Oct 2022 – May 2026
- Graduated with distinction.
- Received merit-based scholarships.

Research

Asking Your Advisor: Learning When to Consult Stronger Models in Agentic Systems
Submitted to ICLR 2027. We train a meta-policy that learns when to consult a stronger advisor model and when to continue with a cheaper one, reaching a better success-cost Pareto frontier than standard advisor-executor setups. Preprint on arXiv.

Programs

TU Wien Informatics, Bachelor with Honors — Vienna, Austria
Oct 2022 – May 2026

- Selected for the Informatics excellence program, a performance-based honors track comparable to the Anglo-American honors model.
- Admission and graduation depend solely on performance, placing graduates among the top 5 percent of students at TU Wien.
- Building an individual advanced curriculum with a mentor, drawing on research-oriented and master's-level courses.

European Student Assembly 2026, EUC Voices, Delegate — Strasbourg, France
Jan 2026 – Apr 2026

- Representing the EULiST alliance at the European Student Assembly 2026.
- Debating key issues and advocating for policy changes at the European level.
- Writing policy recommendations on empowering inclusive digital literacy and building resilience.

Student Engagements

TU Wien Robotics Club, Machine Learning Engineer — Vienna, Austria
Apr 2026 – present

- Implementing vision-language-action models (VLAs) and world models for robotics tasks.
- Leading the reinforcement learning division.
- Building a chess robot that autonomously plays chess and moves the pieces.

European Student Robotics Association (ESRA), Executive Board Member, Partnerships — Europe
Sept 2026 – present

- Serving on the executive board of ESRA, the European Student Robotics Association.
- Driving partnerships to push the boundaries of European robotics.

TU Wien Space Team, Frontend Developer — Vienna, Austria
Mar 2025 – Oct 2025

- Contributing to the "Lamarr" project at the 2025 European Rocketry Challenge (EuRoC).
- Building frontend tooling for rocket system control.

Skills

Programming
Python, Java, C#, HTML, CSS, JavaScript

Machine Learning
Model development and evaluation

Reinforcement Learning
Algorithms, use cases, and applications

Robot Learning
Vision-language-action models (VLAs), world models, applying machine learning to robotic tasks

Software Engineering
Git, Docker, Maven, REST APIs

Human-AI collaboration
shared-initiative interaction, explainable AI, human-in-the-loop

Generative AI
Prompting, API-based integration, prototype development

Languages

German
Native

English
Fluent

Russian
Fluent